

Assignment 3: Customer Development (CustDev)

Report

Course:	Technology				Entrepreneurship		
Project:	SUB-CHECK	—	Cloud-Based	Behavioral	Anti-Cheat	Platform	
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Repository: https://github.com/Scirtor/subcheck-anticheat							

1. Executive Summary

This report documents the Customer Development (CustDev) research conducted for **SUB-CHECK**, a server-side behavioral anti-cheat platform. The core goal of this research was to validate whether the gaming market—specifically competitive players, community tournament organizers, and independent game studios—experiences severe friction with current kernel-level (Ring 0) anti-cheat solutions, and whether an agentic/server-side behavioral telemetry approach addresses their unserved jobs, pains, and gains.

Over a two-week period, our team gathered data from **32 respondents** across three core market segments via structured qualitative questionnaires, supported by **5 in-depth recorded interview sessions**. The findings overwhelmingly validate our foundational problem hypotheses while unveiling critical new insights regarding hardware-based cheating (DMA/Cronus), Linux/SteamOS platform lockouts, and the prohibitive licensing barriers faced by indie multiplayer studios.

2. Target Audience Segmentation & Prioritization

To conduct effective Customer Development, we segmented our potential customer ecosystem into three distinct groups based on accessibility, market volume, and potential satisfaction/impact:

Segment	Description	Accessibility (1-5)	Market Volume (1-5)	Impact / Need (1-5)	Priority
Segment A: Competitive &	Active players in ranked FPS games	5 (High — Discord, Reddit,	5 (Massive — tens of	4 (Directly impacted by	P1

Segment	Description	Accessibility (1-5)	Market Volume (1-5)	Impact / Need (1-5)	Priority
Ranked Gamers	(CS2, Valorant, Apex Legends, R6 Siege) playing >10 hrs/week.	peers)	millions)	cheaters & rootkits)	(Primary)
Segment B: Community Admins & Tournament Hosts	Organizers of collegiate esports, amateur tournament brackets, and dedicated community servers.	4 (Medium-High — university clubs, clan hubs)	3 (Moderate — thousands of organizers)	5 (Critical — manual demo reviews cause burnout)	P2 (Secondary)
Segment C: Indie & Mid-Tier Game Studios	Developers building commercial multiplayer titles without multimillion-dollar R&D budgets.	3 (Moderate — itch.io, indie discords)	3 (Growing segment — thousands of studios)	5 (Extremely high — enterprise AC is unaffordable)	P3 (Strategic B2B)

3. Customer Discovery Methodology

Our interview preparation followed *The Mom Test* principles: we avoided pitching our hypothetical solution during discovery, focused on specific past behavior, and probed deeply into frustrations, workarounds, and economic costs.

- **Jobs to be Done (JTBD):** For players, enjoying fair matches and maintaining system security; for tournament admins, certifying competitive legitimacy quickly; for indie developers, protecting player retention and Steam review scores.
- **Pains Explored:** Frequency of cheater encounters, false positive experiences, intrusive kernel drivers causing BSODs/privacy concerns, hardware cheats (DMA cards, Cronus) bypassing client scans, and OS incompatibility (Linux/SteamOS).
- **Gains Explored:** Transparent risk scoring, zero client installation, seamless cross-platform play, and automated demo analysis.

4. Quantitative & Statistical Findings

Analysis of our 32 structured respondents (recorded in questionnaire/responses.csv and analyzed in analysis/analysis.xlsx) reveals striking patterns:

Perception & Sent

Positive



Figure 1: Player sentiment toward Kernel-level (Ring 0) drivers (62.5% Negative).

ude_Towards_ServerSide_Behavioral

Positive

Neutral

Rec

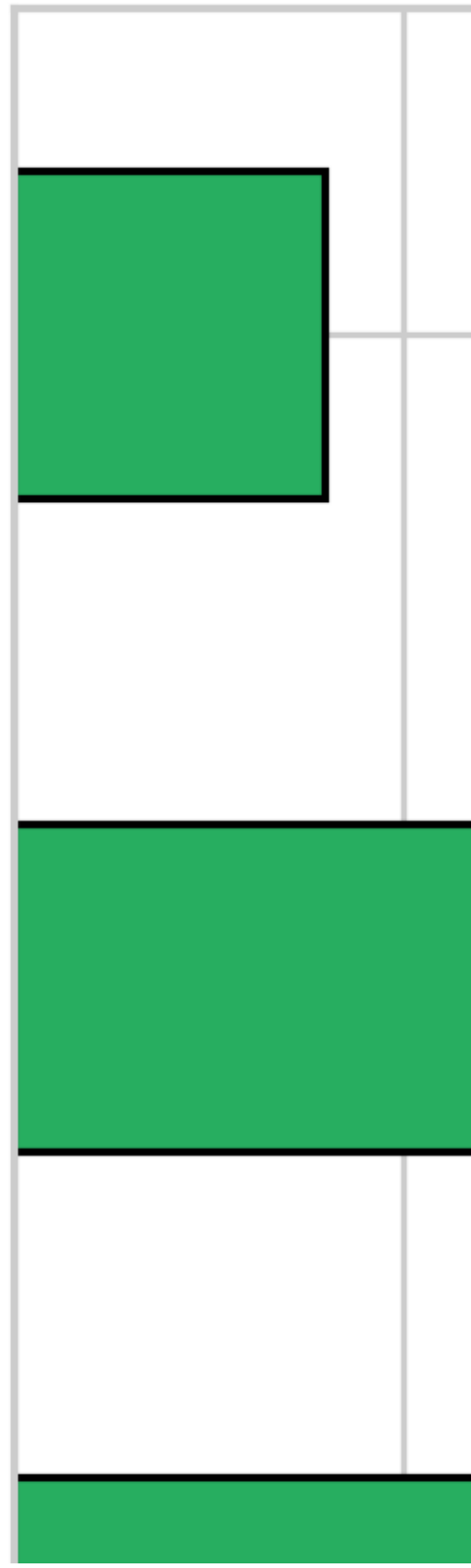


Figure 2: Receptiveness to Server-Side behavioral analytics (93.8% Positive/Extremely Positive).

Reported

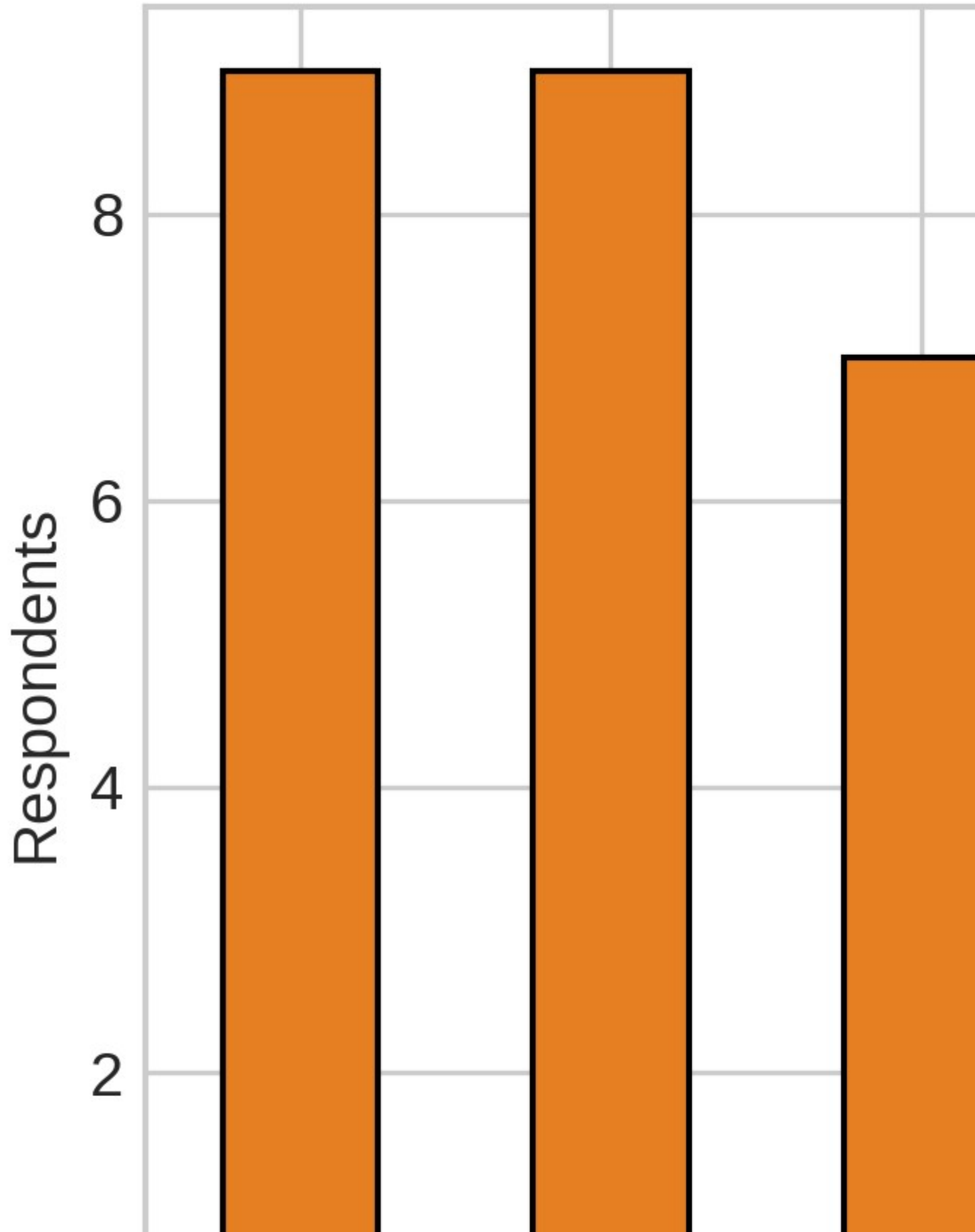


Figure 3: Reported encounter frequency with cheaters in competitive queues.

Willingness to Participate



Figure 4: Willingness to participate in SUB-CHECK pilot testing (87.5% Yes).

5. Key Insights & Customer Quotes

R07 (Esports League Coordinator): "Our biggest nightmare is DMA (Direct Memory Access) cards. Wealthier players buy a second PC with a PCIe FPGA card that reads game memory over hardware. No client-side anti-cheat catches this because the cheat runs on an entirely different machine... A server-side behavioral model is the ONLY way to catch DMA cheats."

R08 (Indie Game Developer): "When we reached out to enterprise anti-cheat vendors like BattlEye and Easy Anti-Cheat, the licensing models and integration overhead were completely unrealistic for our budget (\$10k+ minimum commitment). Small studios are forced to ship unprotected games."

R14 (Linux / Steam Deck Gamer): "Kernel anti-cheats treat Linux users like criminals. I was locked out of my favorite games overnight. A server-side solution doesn't care whether client packets originate from Windows or Linux because it inspects kinematics, not OS memory."

Core Inferences from CustDev:

1. **The Hardware Cheat Blindspot:** Traditional anti-cheats (BattlEye, EAC, Vanguard) rely on OS memory inspection, leaving them structurally blind to DMA hardware cards and external Arduino/Cronus emulators. However, physical mouse kinematics and human jerk limits cannot be forged on the game server.
2. **The Linux / Handheld Opportunity:** Over 15% of respondents actively play on Linux or Steam Deck. Server-side verification creates a massive competitive advantage by offering 100% platform neutrality.
3. **Indie Studio Market Gap:** Current enterprise anti-cheats ignore indie/AA developers due to high minimum licensing fees (\$10k-\$50k). A self-serve SaaS API priced per Monthly Active User (MAU) fills a massive void.

6. Business Idea & Product Updates (Value Proposition Canvas)

Based directly on the customer feedback, we adjusted our initial business model in three critical ways:

Dimension	Initial Business Idea (Pre-CustDev)	Updated Business Model (Post-CustDev)	Rationale from Feedback
Target Customer	Direct-to-consumer gamers via a desktop companion app.	B2B SaaS for Indie/Mid-tier Game Studios & Tournament Hosts.	Gamers cannot mandate anti-cheat adoption; game developers and tournament hosts have the purchasing authority and direct pain.
Pricing Model	Annual fixed enterprise licensing.	Usage-based tiered pricing (\$0.03 – \$0.07 per Monthly Active User) with a free indie tier.	Indie devs strongly demanded predictable, usage-based operational costs rather than upfront six-figure contracts.
Feature Focus	General memory anomaly detection.	Specialized Kinematic Telemetry Engine for DMA & Recoil Macro detection.	DMA cards and hardware emulators were cited by tournament hosts as the #1 unsolvable threat under current tools.
Enforcement Model	Binary automated instant banning.	Risk scoring (0.0 - 1.0) with Quarantine Matchmaking & Admin Dashboard.	Players and developers expressed fear of false positive bans; risk-driven isolation prevents ruining legitimate games while gathering evidence.

7. Conclusion & Next Steps

Assignment 3 validated the commercial and technical thesis for **SUB-CHECK**. The customer feedback confirmed strong market rejection of invasive kernel software and verified the urgent need for a server-side, hardware-agnostic behavioral approach. The team will now proceed to prototype the kinematic feature extraction pipeline and prepare for pilot testing with our identified respondents.